

PROJECT REPORT

Productivity Improvement of 'The Lighthouse' Wind Farm



Bootcamp Data Analytics | Unicorn Academy

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1. PROJECT PURPOSE

This section introduces the central objective of the project and establishes the context guiding the development of the data model and performance evaluation of 'The Lighthouse' Wind Farm.

1.1 Introduction and Project Context

Wind energy has consolidated over recent years as one of the most efficient and reliable technologies within the sustainable generation portfolio. Its capacity to convert an abundant natural resource into electricity on a continuous basis positions it as a competitive alternative, especially in regions with strong and regular winds.

Electricity generation from wind requires a wind generator — also called a wind turbine — which converts the kinetic energy of air into mechanical energy and ultimately into electricity.

These types of projects not only contribute to emissions reduction, but also present relevant technical challenges: climatic variability, power curves, mechanical degradation, and maintenance strategies that directly influence the final performance.

In this context, Aerowind requested a comprehensive performance analysis of The Lighthouse Wind Farm, one of its most representative sites due to its combination of favorable environmental conditions and technological diversity in its turbine fleet.

The wind farm includes turbines installed in 2023, 2024, and 2025. Each has different nominal capacities, making it an ideal case to evaluate efficiency, availability, and comparative operational behavior within the same environment.

In summary, the project aims to deliver to Aerowind a complete diagnostic of The Lighthouse Wind Farm's performance, proposing improvement opportunities based on technical analysis, operational reliability criteria, and solid empirical evidence.

1.2 Project Selection

The selection of a wind farm as a case study was not arbitrary. From my background as a Chemical Engineer and my experience in continuous improvement, data analysis, and industrial processes, the energy industry has always represented an area of particular technical and professional interest.

In particular, wind energy combines three key dimensions for a robust analytical approach:

a. A physically modelable process highly dependent on real data

The energy generated can be contrasted against wind speed, air density, and temperature, allowing deviations to be detected.

b. An operating system with multiple interactions

Availability, failures, maintenance events, and seasonality all influence each turbine's performance, requiring consistent multivariate analysis.

c. An industrial environment that demands data-driven decisions

Managing a wind farm demands precise diagnostics to sustain availability, anticipate failures, and validate real vs. expected performance. Proper analysis reduces uncertainty and enables prioritization of interventions with direct operational impact.

1.3 Wind Farm Description

The Aerowind Wind Farm has characteristics that make it a representative and technically interesting case:

- It is a medium-sized wind farm located in Necochea, Buenos Aires, Argentina.
- It operates in a region with marked seasonal variability, allowing analysis of production sensitivity to weather changes.
- Its location, near the coast and exposed to wind regimes, allows study of whether the installed infrastructure is correctly aligned with the real wind resource potential.

Part of the objective is also to assess whether the patterns observed over the years validate the park's location or whether there are indications that its performance could be improved through operational adjustments or design reconsideration.

This wind farm features:

- 10 wind turbines
- 21,9 MW of installed capacity
- 48 GWh of annual energy generated

1.4 Project Purpose

Aerowind has experienced sustained growth in recent years, incorporating turbines of different models and capacities. However, this development was not accompanied by a unified system to record, analyze, and evaluate its key indicators.

Currently, the company lacks structured information on energy production, operational performance, climatic conditions, and corrective and preventive maintenance, which limits its ability to understand the real performance of each turbine and the farm as a whole.

In practice, Aerowind relies on isolated charts manually created in Excel, without a standardized process for data capture, cleaning, or analysis. This makes it difficult to detect losses, compare turbines, identify trends, and make evidence-based decisions.

The objective of this project is to evaluate the integral performance of The Lighthouse Wind Farm using SQL and Power BI and to develop a comprehensive analytical model that enables:

- a. Consolidating and cleaning the existing database, unifying weather, production, and operational information.
- b. Defining and calculating the main KPIs for the wind farm, including production, efficiency, availability, and performance metrics.
- c. Designing an interactive dashboard that facilitates real-time monitoring of farm performance and provides support for tactical and strategic decisions.
- d. Providing Aerowind with a reliable, scalable, data-driven tool that replaces manual reports and enables more precise and efficient management of its wind operations.

In summary, this project seeks to transform dispersed data into useful information, offering a solid foundation to improve the performance of The Lighthouse Wind Farm and strengthen Aerowind's analytical capacity going forward.

This project analyzes three essential components:

a. Weather

- Wind speed, temperature, pressure, and humidity.
- Identification of seasonality.

b. Energy Production

- Energy generated per turbine, operative and inactive hours.
- Efficiency KPIs: availability, capacity factor, and performance.

c. Maintenance

- Costs, frequency, and type of intervention.
- Relationship between failures and loss of generated energy.

The project includes cleaning and standardizing the available database, correlating key variables, and developing indicators to support technical decision-making.

From this diagnostic, the goal is to have reliable information for strategic decisions aimed at:

- Optimizing energy generation.
- Reducing operational costs associated with maintenance.
- Anticipating critical behaviors of the farm according to environmental conditions.

1.5 Wind Farm Fleet Description

The wind farm began operations in 2023 with the installation of five turbines. In 2024, two additional turbines were incorporated and, finally, in 2025 the expansion was completed with the installation of three new units. Throughout these stages, different turbine models were used, reflecting a progressive technological evolution of the park.

Turbine models installed:



Figure 1. Vestas V100



Figure 2. Vestas V110



Figure 3. GE -2.5 116

Turbine Name	Type	Installation Year	Nominal Capacity
T01, T02, T03, T04, T05	Vestas V100	2023	2,000 kW
T06, T07	Vestas V110	2024	2,200 kW
T08, T09, T10	GE 2.5-116	2025	2,500 kW

1.6 Issues Identified

During the last years of operation of The Lighthouse Wind Farm, Aerowind identified a series of inconsistencies and anomalous behaviors in its productive and operational performance. Despite having turbines with different technologies and very favorable wind conditions, the farm fails to reach the generation levels the company considers expected for its installed capacity.

The main difficulties detected include:

a. Significant production variability among similar turbines

Aerowind observed marked differences in energy generated by units of the same nominal power, without a clear cause associated with wind, maintenance, or reported failures. This raises doubts about the real efficiency of certain turbines.

b. Inconsistencies in operational records

The company has scattered, partial, or incomplete historical data (production, wind, temperature, shutdowns). This lack of standardization prevents accurate evaluation of availability, maintenance events, and the real capacity factor of the farm.

c. Lack of visibility into recurring failures

Some turbines recorded multiple unplanned shutdowns or irregular behaviors in certain periods, but Aerowind does not have a system that allows identification of patterns or root cause determination from the available information.

d. Difficulty correlating weather conditions with production

The company lacks analyses relating wind speed, temperature, or air density to production, making it impossible to know whether the energy generated is consistent with the theoretical power curve of each turbine.

e. Absence of clear indicators to evaluate overall farm efficiency

The lack of unified KPIs (capacity factor, availability, energy losses, downtime hours, curtailments) limits Aerowind's ability to compare performance, estimate losses, and prioritize improvement actions.

1.7 Reason for the Analysis Request

Given these problems, Aerowind requires a technical study that:

- Integrates and cleans all historical data,
- Allows evaluation of generation consistency with respect to weather conditions,
- Identifies efficiency deviations turbine by turbine,
- Delivers reliable information for operational and maintenance decision-making.

This report is developed to address these information gaps, provide evidence-based diagnostics, and offer a clear picture of the real performance of The Lighthouse Wind Farm. It seeks to answer several questions through a structured analysis of actual park data.

2. DATASET DESCRIPTION

This chapter describes the dataset used as the basis for the wind farm analysis. The work was carried out using raw CSV-format tables containing operational, technical, and environmental information about the farm.

Five raw tables (CSV files) were used as the base:

- Production
- Maintenance
- Weather
- Turbines
- Price
- Theoretical Curves

2.1 Production Table

Contains the production records of all wind turbines in The Lighthouse Wind Farm for the years 2023–2025:

Variable	Type	Description
id_record	id	PK – Unique identification of the production record.
id_turbine	id	Unique identification of the wind turbine.
date	date	Production dates.
energy_generated_kwh	decimal	Energy generated in kWh.
operative_hours	decimal	Operative hours of each turbine.
inactive_hours	decimal	Inactive hours of each turbine.
year	integer	Year.
installed_power_kw	integer	Theoretical power according to turbine type (manufacturer) in kW.

2.2 Maintenance Table

Contains the maintenance activity records for all turbines in the The Lighthouse Wind Farm for the years 2023–2025:

Variable	Type	Description
id_maint	id	PK – Unique identification of the maintenance record.
id_turbine	id	Unique identification of the wind turbine.
maint_date	date	Maintenance dates.
maintenance_type	categorical	Type of maintenance.
duration_hs	decimal	Maintenance duration in hours.
cost_usd	decimal	Maintenance cost in USD.

2.3 Weather Table

Contains records of weather variables at The Lighthouse Wind Farm location for the years 2023–2025, obtained from the Meteorological Station (MET MAST) installed at the park:

Variable	Type	Description
date	date	Date of weather variable recording.
wind_speed_mps	decimal	Wind speed in m/s.
temperature_c	decimal	Air temperature in °C.
relative_humidity_pct	decimal	Percentage of relative humidity of the air.
pressure_hpa	decimal	Atmospheric pressure in hPa.

2.4 Turbines Table

Contains details of turbines installed between 2023–2025:

Variable	Type	Description
id_turbine	id	PK – Unique identification of the wind turbine.
model	text	Turbine model.
location	text	Turbine location (Wind Farm).
nominal_power_kw	integer	Nominal/theoretical capacity in kW.
installation_date	date	Turbine installation date.

2.5 Prices Table

The prices table contains the sale value of electrical energy expressed in USD/kWh for each period. This information is used as a key input for calculating income and profitability of the wind farm, enabling conversion of generated energy (kWh) into comparable economic results across years and months.

Variable	Type	Description
id_price	id	PK – Unique identification of monthly price.
date	date	Monthly date.
price_kwh_usd	decimal	Sale price of generated energy per month (USD/kWh).

2.6 Theoretical Curves Table

This table describes the expected performance of each turbine model at different wind speeds, establishing the theoretical relationship between wind speed and generated power. This information represents the nominal design performance of the turbines, without considering operational losses, shutdowns, or real operating conditions.

The table is used as a reference to compare actual generated power with theoretical available power, enabling evaluation of the degree of wind resource utilization, identification of performance deviations, and analysis of energy efficiency of each turbine model within the farm.

Variable	Type	Description
model	text	Turbine model.
speed_mps	integer	Wind speed in m/s.
theoretical_power_kw	integer	Theoretical power per model according to wind speed.

In particular, the theoretical curves table does not represent a traditional business dimension, but rather a technical reference dimension, used to describe the theoretical behavior of the system and to allow comparisons between real generation and available energy potential. Its function is to contextualize the facts from a physical standpoint rather than a categorical one.

3. DATA CLEANING

Before proceeding with the exploratory analysis and the construction of the relational model, a comprehensive review of the data schema was performed. The goal was to ensure that all tables met minimum criteria of structure, consistency, and inter-table compatibility, thus enabling reliable subsequent modeling and precise indicator creation.

This stage included validation of column formats, verification of data types, control of keys needed for relationships, and standardization of nomenclature across tables.

3.1 Environments, Languages and Tools

SQL was used for initial quantitative and qualitative validation, while Power BI was used for visual analysis.

Environments:

Environment	Use
MySQL Workbench	Validation, cleaning, and technical queries
Power BI Desktop	Relational modeling, DAX measures, and executive dashboard

Languages:

Language	Description
SQL	Language for querying and managing relational databases.
DAX	Expression language used in Power BI to create calculations on the tabular model.

3.2 Schema Transformation and Validation in SQL

The first step of the project consisted of carrying out a complete audit and sanitization process of the aerowind relational database, which concentrates the operational information of the wind farm: meteorological conditions, production records, maintenance events, and technical characteristics of the turbines.

Since the energy analysis and Power BI models strongly depend on the temporal and physical consistency of the data, an exhaustive SQL cleaning was executed to guarantee integrity, traceability, and technical coherence.

3.3 Initial Diagnosis and Database Structure

Below is the preliminary diagnostic of the weather, production, maintenance, turbines, price, and theoretical_curves tables to evaluate temporal coherence, valid ranges, and general structure. This diagnostic allowed establishing the baseline and directing subsequent cleaning criteria.

Once the raw CSV files were loaded into MySQL, a quantitative and qualitative survey of the table status was conducted.

The objective was to establish:

- Initial record volume.
- Existence of duplicates.
- Presence of critical nulls.
- Values outside physically possible ranges.
- Consistency between tables that should be related by natural keys (date → weather/production, id_turbine → production/maintenance).

This diagnostic allowed dimensioning the level of intervention required before advancing to modeling.

3.4 Date Standardization

Raw tables (RAW) corresponding to each data source were defined: turbines, weather, production, maintenance, energy prices, and theoretical curves. These tables were designed with flexible data types (mainly VARCHAR for dates) to facilitate the initial import of CSV files without loss of information.

Once the data was imported, date normalization was performed, converting text-type fields to DATE format using conversion functions, and then modifying the data type of the corresponding columns.

3.5 Duplicate Identification and Treatment

In order to ensure the temporal integrity of the data and avoid distortions in KPIs, a systematic duplicate analysis was carried out on those field combinations that, by model design, should be unique.

Uniqueness was verified in the following tables:

- Weather: one row per date
- Production: one row per date and id_turbine
- Maintenance: one row per id_turbine and maint_date
- Turbines: one row per id_turbine
- Prices: one row per date

- Theoretical curves: one row per wind speed and turbine model

As a result of the analysis, duplicates were only detected in the production table. Multiple records with the same date and id_turbine combination were found, which is incompatible with the model's target granularity: one daily record per turbine.

Analysis of these duplicates revealed that they corresponded to partial measurements taken at different times of the day (up to three daily records for the same turbine). Since these values do not represent inconsistencies but rather temporal fragmentations of the same day, a daily consolidation process was performed.

Records were grouped by id_turbine and date, applying the following aggregation rules:

- Operational variables: consolidated using sum, to reflect the total daily value.
- Environmental variables: consolidated using average, as they represent representative measurements of environmental conditions.

This way, partial records were unified into a single daily row per turbine, preserving temporal coherence of the dataset and ensuring consistency for indicator calculation. An intermediate table (production_daily_base) and a final table (production_daily) with a consistent daily identifier were generated.

3.6 Null and Physically Meaningless Value Verification

Below, critical nulls and values violating physical or operational constraints were evaluated.

3.6.1 Production Table

a. Null or empty values: None detected.

b. Invalid energy: energy_generated_kwh < 0 or IS NULL.

Note: Null and negative values were searched but not zero values, since an energy value of zero is useful for detecting turbine failures.

A zero value indicates a recorded operational event — the turbine was running (with operative hours recorded) but produced no energy due to external conditions (lack of wind) or specific operational failures.

A NULL value represents absence of data or a measurement/loading error. Result: No invalid energy data detected.

c. Negative energy: No negative energy records detected.

d. Physical operational hours: operative_hours < 0 or > 24; inactive_hours < 0 or > 24; operative_hours + inactive_hours > 24. Result: No out-of-range records detected.

3.6.2 Maintenance Table

- a. Null values detected in the category field for preventive maintenance records.

Result: Records where the category field was null for Preventive maintenance type were detected. This does not correspond to a loading error but to a historical system practice: preventive maintenance does not have an associated failure category. These null values were replaced with the standardized category 'Preventive – No Failure'.

- b. Invalid costs and durations: cost_usd IS NULL or ≤ 0 ; duration_hs IS NULL or ≤ 0 .
Result: Records with negative costs were deleted as loading errors. Null or zero durations were also deleted, as they do not represent valid maintenance tasks.

3.6.3 Weather Table

- a. No null or empty values detected.
- b. Out-of-range temperatures ($< -20^{\circ}\text{C}$ or $> 60^{\circ}\text{C}$): Values outside this range were detected and deleted, as they represent instrumental or loading errors.
- c. Invalid wind speed (< 0 or IS NULL): Erroneous records deleted.

3.6.4 Turbines Table

- a. No null or empty values detected.
- b. Valid nominal power: One turbine was identified with a nominal power value below zero, which is not coherent with real factory specifications. Since the model and installation year were known, the value was standardized by assigning the corresponding nominal power for the detected case.
- c. Installation dates out of range: No out-of-range installation dates detected.

3.6.5 Prices Table

- a. No null or empty values detected.
- b. Invalid price (price_kwh_usd ≤ 0): Invalid price detected and deleted.

3.6.6 Theoretical Curves Table

- a. No null or empty values detected.
- b. Negative power or speed: No negative powers or speeds detected.

3.7 Cross-Validation and Inter-Table Consistency

Additionally, expected cardinalities between tables were validated prior to tabular modeling in Power BI. This step was key to ensure the relational structure correctly reflects the operational business logic and that no ambiguous or inconsistent relationships exist.

a. Production associated with non-existent turbines: A referential integrity validation was performed between the production and turbines tables. Result: No production records with non-existent turbines detected.

b. Maintenance associated with non-existent turbines: The same referential integrity validation was applied to the maintenance table. Result: Maintenance records whose `id_turbine` had no correspondence in the turbines table were detected and deleted to preserve data model coherence.

c. Production prior to turbine installation: No turbine can generate energy before it enters service. Result: 20 inconsistent records were identified and deleted.

d. Maintenance prior to turbine installation: Analogously, no technical intervention can exist on equipment not yet installed. Result: 25 inconsistent records were detected and deleted.

e. Weather outside the temporal range of production: All dates recorded in the weather table fall within the temporal range covered by production records. Result: No inconsistencies detected.

3.8 Creation of Clean Tables and Final Model Load

Once all validations, corrections, and deletions on the raw tables were completed, clean tables were created. These consolidate only valid, consistent, and normalized records, ensuring referential integrity, temporal coherence, and data quality. Subsequently, the final model was loaded from these clean tables, establishing an optimized structure for analysis and indicator generation.

4. DATA MODELING

An extended star architecture — or constellation model — was used for this analytical model, involving multiple operational processes.

In this case, the wind farm analysis requires integrating production, maintenance, weather, and technical characteristics data, which demands more than one fact table and shared dimensions. This structure enables a complete analysis of energy performance and the factors influencing operations.

4.1 Table Types

The model consists of:

Fact Tables:

- Production: concentrates records of energy generated and operative hours of each turbine.
- Maintenance: gathers corrective and preventive interventions with their dates, durations, and costs.

Dimension Tables:

- Turbines: records turbines and their installation date.
- Weather: provides relevant meteorological variables to interpret energy performance.
- Price: records the value of energy generated per month.
- Theoretical Curves: includes theoretical power per wind speed for each turbine model.
- Calendar: table generated in Power BI to enable temporal analyses and historical comparisons.

4.2 Table Relationships

The model is structured in Power BI under a constellation architecture, where fact tables relate to different dimensions through primary and foreign keys. All relationships follow 1 → * cardinality, ensuring traceability, integrity, and consistent filter flow in the analysis.

Although tables are conceptually related through key fields, relationships were not physically implemented in MySQL, as that layer was used primarily for cleaning, normalization, and data consolidation. Relationships were defined in Power BI, where the semantic model was built, allowing greater control over filter direction and metric construction.

a. Relationships of the 'production' fact table:

Fact	Dimension	Relationship	Description
production	weather	date 1 → *	Links each generation record with meteorological conditions.
production	turbines	id_turbine 1 → *	Associates each energy generated record with the corresponding turbine.
production	calendar	date 1 → *	Enables temporal analyses (day, month, year, seasonality).

b. Relationships of the 'maintenance' fact table:

Fact	Dimension	Relationship	Description
maintenance	turbines	id_turbine 1 → *	Allows identification of which turbine was serviced and analysis of operational impact.
maintenance	calendar	maint_date 1 → *	Facilitates temporal analysis of failures, shutdowns, and preventive tasks.

c. Calendar table creation and relationships:

A dedicated calendar table was created to standardize all temporal variables in the model. By generating a continuous calendar without gaps from the first to the last day of the analyzed period, all time-dependent tables relate correctly.

Related Table	Column in Table	Column in Calendar	Relationship Type
weather	date	Date	1 → *
maintenance	maint_date	Date	1 → *
production	date	Date	1 → *

4.3 Power BI Model Overview

The model was organized around two fact tables:

- **Production:** consolidates generated energy, operating and downtime hours, and enables both time-based and per-turbine analysis.
- **Maintenance:** records work orders, duration, and cost, supporting availability and failure analysis.

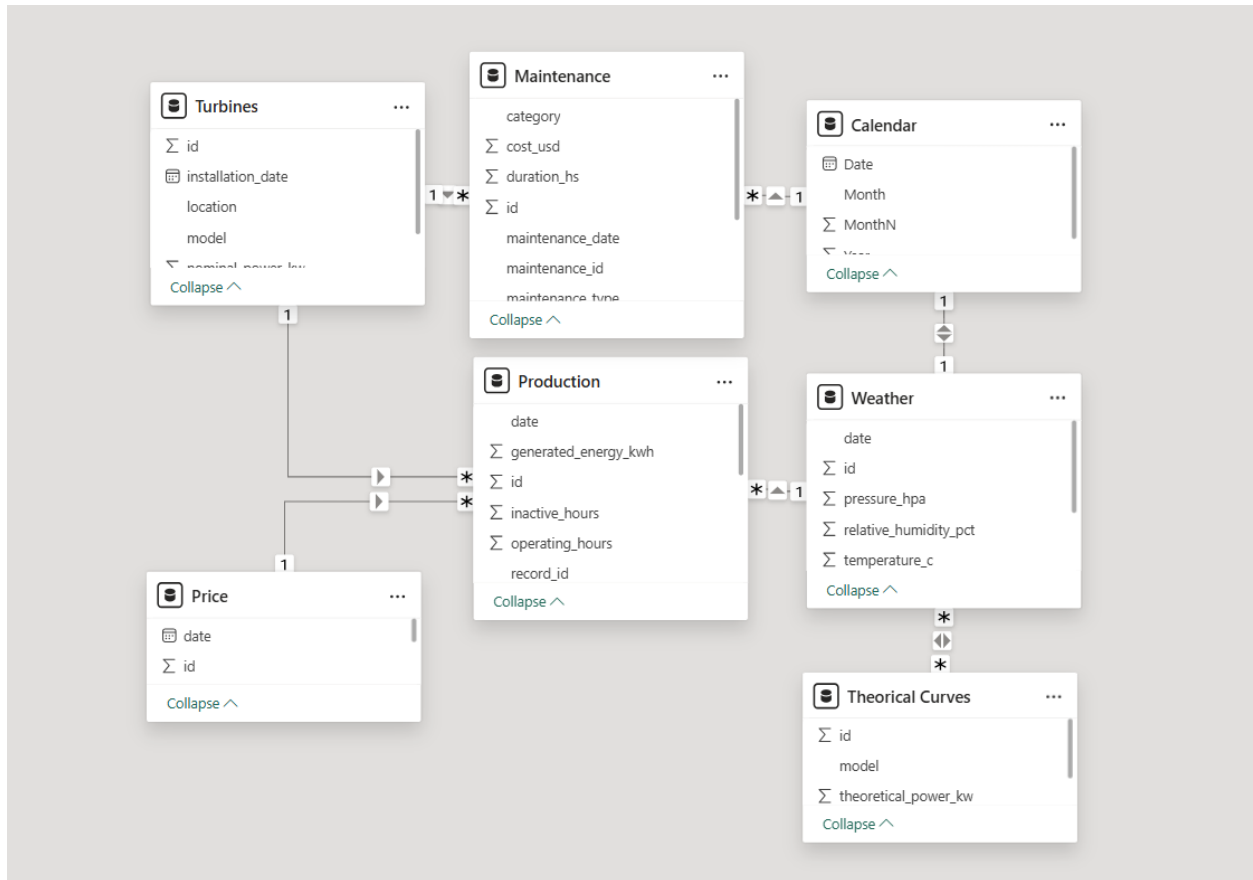


Figure 4. Dimensional relationship between tables.

Both tables relate to dimension tables covering turbines, weather, pricing, and calendar, along with a technical reference table (theoretical curves). Together, they form an analytical schema that preserves referential integrity and enables the joint evaluation of generation output, weather conditions, and operational behavior per turbine.

5. DAX MEASURES AND KEY VISUALIZATIONS

The development of the analytical model in Power BI required creating a set of DAX measures specifically designed to answer business questions, integrate information from multiple tables, and enable visualizations consistent with the real behavior of the wind farm turbines.

5.1 DAX Measures Created

a. Total Energy (GWh and kWh)

Measures total farm production. This is the base metric for evaluating operational performance, comparing periods, and studying the relationship with weather.

b. Real Power (kW)

Represents the electrical power effectively generated by the turbine under real operating conditions, considering wind behavior, mechanical and electrical losses, and limitations imposed by control systems.

c. Theoretical Power (kW)

Corresponds to the maximum estimated power available in the wind for a given speed, calculated from the physical wind resource model and the geometric characteristics of the rotor, without considering losses or operational restrictions.

d. Availability (%)

Evaluates what percentage of time the turbine was operative relative to the total day, meaning it was not stopped for maintenance, failures, or unavailability. Allows quantification of the real productivity of the farm.

Availability = Operative Hours / Total Hours

e. Capacity Factor (%)

The capacity factor represents the turbine's efficiency, meaning the energy conversion efficiency of the wind farm. This metric compares the energy actually generated against the maximum theoretical energy possible if all turbines operated at rated power for every hour of the period.

Formally, it is defined as:

$$CF = \frac{\text{Real energy production (kWh)}}{\text{Maximum possible energy production (kWh)}}$$

Which is equivalent to:

$$CF = \frac{\text{Real energy production (kWh)}}{\text{Rated power (kW)} \times \text{Hours in the period (h)}}$$

The capacity factor is a key metric indicating how efficiently a wind turbine operates over time. It is defined as the ratio between the actual electrical energy produced during a specific period and the maximum energy it could have produced had it operated continuously at rated or full capacity throughout that same period.

Maximum possible production is calculated by multiplying the turbine's rated capacity (peak power) by the total number of hours in the period (e.g. 8,760 hours in a year). It reflects how efficiently the wind farm operates relative to its theoretical potential.

Typical capacity factors for onshore wind farms range between 20–35%, depending on wind resource quality and turbine technology mix.

f. Performance (%)

Performance is the main integrating metric of the analysis, combining in a single indicator the key factors that define the global efficiency of each turbine. It is calculated as a weighted combination of availability, capacity factor, and a penalty for corrective maintenance events:

$$\text{Performance} = 0.65 \cdot D + 0.30 \cdot F + 0.05 \cdot P$$

Where:

- D = Availability,
- F = Capacity Factor,
- P = Penalty for corrective maintenance count.

Performance ranges:

- **High performance (80-100):** outstanding performance / optimal operation
- **Medium (60-80):** acceptable operation with room for improvement
- **Low (<60):** below-expected performance / requires review

g. Total Maintenance Cost (USD)

Evaluates the economic effort invested in sustaining operations.

h. Number of Maintenance Events

Enables counting of maintenance interventions, both corrective and preventive.

5.2 Power BI Visualizations

5.2.1 'Energy Performance' Dashboard

This dashboard evaluates the energy performance of the farm. It integrates the three key operating indicators of the wind farm into a single panel, enabling analysis of monthly behavior, historical annual trends across 2023, 2024, and 2025, and identification of trends, seasonality, and deviations from targets.

Content:

- Monthly and annual capacity factor (%): seasonal and inter-annual evolution of wind resource utilization.
- Monthly and annual availability (%): tracking of farm operational continuity.
- Monthly and annual performance (%): key indicator with traffic light color coding.

5.2.2 'Operational Reliability' Dashboard

This dashboard analyzes maintenance performance of the farm, evaluating frequency, type, and cost of interventions per turbine and per period, and their relationship with operational availability. The focus is on identifying critical equipment, failure patterns, and technical and economic optimization opportunities.

Content:

- Monthly maintenance events: monthly evolution of interventions, differentiating corrective and preventive.
- Maintenance per turbine: comparison of intervention volume per turbine.
- Cost per turbine (USD): distribution of total maintenance cost per equipment and intervention type.
- Maintenance history: annual balance between corrective and preventive.
- Top failure types: most frequent failures impacting operations.
- Availability vs. cost correlation: analysis of the trade-off between operational performance and maintenance expenditure per turbine.

5.2.3 'Wind Resource' Dashboard

This dashboard analyzes the influence of wind speed on energy generation and turbine power. The focus is on understanding the physical-operational relationship between wind resource and real farm performance. It enables validation of expected behavior against theoretical models, detection of improvement opportunities in wind utilization, and explanation of energy performance from a climatic and technical perspective.

Content:

- Wind vs. generated energy scatter: direct relationship between wind speed and energy produced.
- Wind speed and generated energy (monthly): joint evolution identifying wind resource seasonality.
- Real power vs. wind speed scatter: comparison of each turbine's behavior against wind, detecting deviations or performance differences.
- Real power vs. theoretical power: contrast between the park's real curve and the expected theoretical curve.

5.2.4 'Profitability' Dashboard

This dashboard evaluates the economic result of the farm, integrating energy generation, income from energy sales, operational costs, and profit, both monthly and annually.

Content:

- Income from energy sold (USD): monthly evolution of generation-associated income.
- Costs (USD): monthly maintenance costs of the farm.
- Energy generated (kWh / GWh): monthly and total energy production.
- Profit (USD): monthly and cumulative annual economic result.

5.2.5 'Daily Turbine Monitoring' Dashboard

This dashboard presents daily evolution charts of energy generated per turbine for the last available year. It is used as an operational control panel for daily performance tracking of each turbine. Its main objective is to early detect operational anomalies, abrupt production drops, or performance deviations from expected behavior.

Content: One chart per turbine showing the daily evolution of energy generated.

6. EXPLORATORY ANALYSIS

The exploratory analysis constitutes the central stage of the project, where wind farm data is systematically examined to understand real operational behavior, validate consistency across tables, and reveal operational patterns. This initial analysis enabled the identification of trends, detection of anomalies, and provided the foundation for building operational KPIs. The following sections describe each visual and its results, organized by Power BI tab.

6.1 Scope and Variables Analyzed

The analysis was developed for the period 2023–2025, considering different temporal aggregation levels (monthly and annual) and operational disaggregation levels (per turbine).

Main variables analyzed:

- Energy generated (kWh / GWh)
- Capacity factor
- Operational availability
- Wind speed
- Real and theoretical power
- Number and type of maintenance events
- Operational costs, income, and profit

6.2 Energy Performance

6.2.1 Analysis

A favorable evolution of farm performance is observed over the 2023–2025 period. Energy indicators show progressive annual improvement, with sustained increases in both the capacity factor and the overall system performance.

In 2023, the farm presents the lowest values of the analyzed period, with moderate energy performance and marked monthly variability. Indicators reflect initial resource utilization, with peaks concentrated in the middle months of the year and pronounced drops toward the edges of the period.

During 2024, improvement is observed over the previous year, with an increase in annual capacity factor and aggregate performance. While the seasonality observed in 2023 is maintained, monthly values show greater consistency and reduced variability amplitude, indicating positive evolution of overall farm performance.

In 2025, the last year considered, the farm achieves the highest capacity factor and annual performance values of the analyzed period. Seasonality continues, with peaks in the central months, but the general level of indicators is above previous years. Operational availability remains stable, supporting energy performance improvement without evidencing deterioration in operational continuity.

6.2.2 Key Findings

- Progressive improvement in annual energy performance between 2023 and 2025.
- Performance seasonality remains constant across all three years, with peaks in the central months.
- Operational availability shows stable values with limited inter-annual variations.
- 2025 is the year with the best aggregate energy performance within the analyzed horizon.

6.2.3 Business Questions Addressed

Q1. Is the energy performance improvement sufficient to justify overall farm performance against installed capacity?

Analysis

The improvement is consistent with better utilization of installed capacity. The increase in capacity factor and global performance occurs without significant increases in operational availability, which remains practically constant. This indicates the farm generates more energy per installed MW and per available hour, ruling out that improvement comes from greater operation time.

Conclusion

Yes. The energy performance improvement justifies the overall farm performance against its installed capacity, evidencing more efficient asset utilization without requiring increases in installed power or operational availability.

Q2. Is the increase in capacity factor homogeneous throughout the year or concentrated in specific periods?

Analysis

The increase in capacity factor is not homogeneously distributed throughout the year, but responds to a structural seasonal pattern with peaks concentrated in the winter months. This is consistent with the indicator's definition and with the physics of the wind resource.

The theoretical power available in the wind follows the physical formula:

$$P = \frac{1}{2} \cdot \rho \cdot A \cdot w^3$$

Where:

P is power

ρ is air density

A is the rotor swept area

w is wind speed.

Since power is proportional to the cube of wind speed, relatively small increases in average winter speed generate significant increases in energy produced, raising the capacity factor. In summer months, even with similar availability levels, lower average wind speed limits achievable power, reducing the indicator's value.

Conclusion

The capacity factor increase is conditioned by resource seasonality with dominant winter peaks, but the improvement observed in 2025 reflects better wind utilization throughout the entire year, elevating the farm's global energy performance.

Q3. Can the energy performance achieved in 2025 be considered a new baseline for the farm or an isolated peak?

Analysis

The 2025 energy performance presents consistently throughout the year, maintaining the park's characteristic seasonal pattern without evidencing isolated peaks or atypical behaviors. The improvement builds on the trend already observed in 2024, not as an abrupt jump.

Conclusion

The 2025 energy performance can be considered a new baseline of farm performance and not an isolated peak, being representative of expected performance under normal operating conditions.

Q4. Did the new technology (V110 and GE 2.5-116) economically improve the farm?

Analysis

In 2023, the farm operated exclusively with Vestas V100 turbines, presenting an annual capacity factor of approximately 18%. From 2024 with V110 incorporation, and 2025 with GE 2.5-116 (T008–T010), progressive improvement in energy performance was observed. In 2025, the annual capacity factor reaches values close to 28%.

Year	Technology	Annual Capacity Factor
2023 (V100 only)	Vestas V100	≈ 18%
2024 (with V110)	Vestas V110 added	≈ 21%
2025 (with GE)	GE 2.5-116 added	≈ 28%

Conclusion

Yes. Incorporating V110 and GE 2.5-116 turbines increased the farm's capacity factor by ~10 percentage points (from 18% to 28%), with similar availability maintained (~90%). Data justifies continued investment in larger rotor and higher nominal power models, especially the GE 2.5-116 family.

6.3 Reliability

6.3.1 Analysis

The reliability analysis shows differentiated evolution in the maintenance behavior of the wind farm over 2023–2025, both in number of interventions and associated costs.

In 2023, the farm presents a low number of maintenance events, concentrated in few turbines, with reduced total cost. Corrective maintenance predominates, but the overall economic impact is limited, reflecting a contained operational level in terms of intervention.

During 2024, a significant increase in maintenance quantity and total cost is recorded, with greater dispersion among turbines. Corrective maintenance continues to represent a relevant proportion, and cost per turbine shows greater heterogeneity.

In 2025, while the total number of maintenance events continues to increase, total cost decreases compared to 2024. A shift in the composition of interventions is observed, with greater participation of preventive maintenance and more balanced cost distribution among turbines. This suggests improvement in the maintenance scheme efficiency.

The correlation analysis between availability and maintenance cost shows that turbines with higher costs do not necessarily present lower availability levels, indicating that increased maintenance spending does not always translate into direct operational continuity penalties.

6.3.2 Key Findings

- The number of maintenance events progressively increases between 2023 and 2025.
- The cost peak is recorded in 2024, while in 2025 costs decrease despite a higher number of interventions.
- In 2025, a shift in the maintenance mix is observed, with greater preventive vs. corrective weight.

- Maintenance costs show concentration in specific turbines, especially in 2024.
- No direct linear relationship is evidenced between higher maintenance cost and lower availability.

6.3.3 Business Questions Addressed

Q1. Does maintenance cost evolve proportionally to the number of interventions?

Analysis

The year-over-year comparison indicates that total maintenance cost does not scale proportionally with the number of interventions. In 2025, total cost decreases relative to 2024 despite a higher number of maintenance events.

This is attributable to a structural shift in the maintenance mix toward preventive interventions, which carry lower unit costs than corrective ones. Corrective maintenance typically entails longer repair times, unplanned spare parts consumption, and unscheduled logistics, all of which drive up cost per event. As preventive activity increases, unplanned failures are reduced, lowering both the frequency and average cost of corrective interventions.

Conclusion

No. Maintenance cost does not scale linearly with intervention volume. The increased weight of preventive maintenance in 2025 reduced the incidence of high-cost corrective events, resulting in lower total expenditure despite greater overall maintenance activity.

Q2. Are we spending on maintenance where it truly impacts farm reliability?

Analysis

Maintenance costs are not homogeneously distributed — they are concentrated in a small subset of turbines. T003 and T001 appear consistently as the highest cost units across all three years. Despite this concentration, no proportional drop in operational availability associated with these turbines is observed, suggesting that a significant part of the maintenance performed contributes to sustaining farm reliability. In contrast, newer turbines T008, T009, and T010 show low costs and minimal economic impact.

Conclusion

Yes. Maintenance spending is focused on specific turbines that concentrate the greatest economic and operational impact. The recurrence of T003 and T001 as cost focal points justifies targeted reliability management over these units.

Q3. Are there turbines with recurring corrective failure patterns impacting farm reliability?

Analysis

Turbine T003 presents a recurring pattern of corrective failures across all analyzed years, with recurrence year after year. The Pareto analysis of failure types shows that

the largest proportion of corrective events in T003 is associated with overtemperature shutdowns. This repetition suggests a structural condition of the asset, possibly related to the cooling system or aging of temperature-sensitive components.

Conclusion

Yes. Turbine T003 has a recurring corrective failure pattern that directly impacts farm reliability and economic performance. This pattern justifies prioritizing it as a critical asset, focusing predictive maintenance actions and specific technical analysis on the dominant failure modes identified.

6.4 Wind Resource

6.4.1 Analysis

The wind resource analysis for 2023–2025 evaluates how wind variability conditions the farm's energy generation and to what extent performance improvements respond to the available resource or to system efficiency.

Across all three years, a positive relationship between wind speed and generated energy is observed, consistent with expected physical behavior. However, point dispersion shows that for the same wind speed, energy produced is not uniform, indicating the influence of additional factors such as turbine technology, aerodynamic efficiency, and operating conditions.

In 2023, moderate average wind speeds and wide dispersion are observed. During 2024, a similar velocity pattern is maintained but with improved conversion efficiency. In 2025, the wind resource retains comparable seasonal behavior to previous years, with winter maximums, but for similar speeds, generated energy is greater and more stable.

Overall, the exploratory analysis shows that the improvement in farm energy performance is not explained by a substantial increase in wind resource, but by more efficient utilization of available wind, particularly visible in 2025.

6.4.2 Key Findings

- The relationship between wind speed and generated energy is positive across all three years.
- The wind regime maintains a stable seasonal pattern with winter maximums.
- For similar wind speeds, generated energy progressively increases from 2023 to 2025.
- In 2025, lower dispersion between real and theoretical power is observed, indicating greater aerodynamic and operational efficiency.
- Energy performance improvement is not associated with greater wind resource but with better utilization of it.

6.4.3 Business Questions Addressed

Q1. Is the energy generation improvement explained by greater wind resource or greater farm efficiency?

Analysis

Comparative analysis of average wind speeds confirms that the available wind resource remains relatively stable throughout the 2023–2025 period, with no significant structural change in the seasonal regime. However, energy output for equivalent wind speed ranges increases progressively — most markedly in 2025.

The narrowing gap between actual and theoretical power, combined with greater data concentration in the wind–energy scatter, indicates that the park converts wind into electricity more efficiently over time, independent of any meaningful increase in available resource.

Conclusion

The energy generation improvement is mainly explained by greater efficiency in wind resource utilization, not by a substantial increase in available wind speed.

Q2. Does the farm uniformly harness wind across the entire speed range?

Analysis

Analysis of actual versus theoretical power as a function of wind speed demonstrates that resource utilization is not uniform across the operational range.

At low wind speeds, energy output is constrained by physical and operational limitations inherent to the turbines, including cut-in speed thresholds and mechanical and electrical losses, which reduce conversion efficiency. In the medium speed range, actual power progressively converges with theoretical power, representing the zone of maximum operational efficiency. At high wind speeds, output stabilizes rather than continuing to increase, as turbine control systems (pitch regulation and torque control) cap power extraction to prevent structural and electrical overload.

This behavior defines a physical saturation threshold beyond which additional wind speed increments do not yield proportional energy gains.

Conclusion

No. The farm does not uniformly harness wind resource across the entire speed range. A saturation limit exists determined by turbine design. This behavior is consistent with expected system functioning and the improvement is concentrated in the medium speed range.

Q3. Which wind speed range explains most of the park's annual generated energy?

Analysis

Joint analysis of the wind speed frequency distribution and the power curve indicates that the medium wind speed range accounts for the dominant share of annual energy

output. This range combines a high frequency of occurrence with near-peak conversion efficiency, maximizing the contribution to total generation.

High wind speeds, while yielding greater instantaneous power, occur less frequently and are subject to rated power limitations that cap their contribution. Low wind speeds, though frequent, generate little energy due to low conversion efficiency near the cut-in threshold.

Conclusion

Most of the park's annual energy is explained by performance in medium wind speeds, where occurrence frequency is high and system conversion efficiency is maximum. Any improvement in capture, control, or availability within this range has a significantly greater impact than actions focused on wind regime extremes.

Q4. What implications does wind resource behavior have for future expansions or repowering?

Analysis

Given that annual energy generation is predominantly driven by performance at medium wind speeds, and that the park already operates near its saturation limit at high speeds, future capacity additions should prioritize turbine technologies optimized for efficient conversion in the medium speed range. Increasing rated power for extreme wind conditions offers limited incremental benefit relative to the park's actual resource profile.

Conclusion

Expansion or repowering decisions should be oriented toward technologies with greater aerodynamic efficiency and better performance at medium speeds, aligned with the site's real regime.

6.5 Profitability

6.5.1 Analysis

The profitability analysis shows sustained improvement in the wind farm's economic result over the 2023–2025 period, driven mainly by progressive increases in energy generated and associated energy sales income.

In 2023, the farm presents the lowest annual energy and income levels, with a positive but limited economic result. During 2024, both total energy generated and annual income increase, accompanied by higher operational costs. The income growth surpasses cost growth, resulting in improved annual profit. In 2025, the farm achieves its best economic performance of the analyzed period, with the highest energy generated, income, and annual profit.

6.5.2 Key Findings

- Farm profitability improves sustainably between 2023 and 2025.

- Annual profit increase is strongly associated with increased energy generated.
- Income grows at a faster rate than operational costs.
- Generation seasonality directly translates into income and profit seasonality.
- 2025 is the year of greatest economic profitability for the farm.

6.5.3 Business Questions Addressed

Q1. Is the farm's profit growth driven primarily by higher income or cost reduction?

Analysis

Year over year analysis indicates that profit growth reflects both factors: higher revenues and improved cost efficiency operating simultaneously.

Between 2024 and 2025, generated energy and associated revenues increase substantially following the addition of new turbines. In parallel, total operating costs remain stable or decline, despite the expanded fleet and higher maintenance activity. This outcome reflects a structural improvement in economic efficiency, where productive scale expands without a proportional increase in cost.

Conclusion

Farm profit growth is driven by both higher income from greater energy generation and improved cost efficiency. In 2025, the farm generates more energy and operates with more installed turbines while even reducing total cost levels (fewer corrective maintenance events), evidencing clear economies of scale capture and more efficient operational management.

Q2. Does wind resource seasonality directly impact monthly farm profitability?

Analysis

The seasonal pattern observed in energy generation is directly reproduced in monthly revenue and profit figures. Months with higher wind resource and greater generation concentrate the majority of the annual economic result, while lower-generation months yield reduced profitability.

Importantly, however, no month within the analyzed period records a net operating loss. Even during periods of reduced generation, revenues consistently exceed operating costs, indicating a cost structure sufficiently flexible to absorb seasonal variability and an underlying business model robust enough to sustain profitability year-round.

Conclusion

Yes. Farm profitability is seasonal and directly influenced by wind resource availability. However, operations remain profitable throughout the year, with no months showing negative economic results, evidencing high economic resilience of the farm against seasonal wind variability.

7. INSIGHTS AND CONCLUSIONS

The integrated analysis of energy performance, reliability, wind resource, and profitability enabled identification of consistent patterns in farm behavior and the impact of technological and operational decisions.

7.1 General Conclusions

The analyzed wind farm shows a positive evolution in energy, operational, and economic performance over the 2023–2025 period. The increase in capacity factor, greater energy generated, and improved economic results evidence more globally efficient performance, particularly in the last year of the evaluated period.

Results show that this evolution is strongly associated with the progressive incorporation of new turbine technologies and more efficient farm operations, maintaining stable availability levels and a controlled cost structure, even with an increased installed fleet.

In contrast, turbines with operational and economic performance below park average were identified, with T003 standing out as the main critical asset, and to a lesser extent T001 and T004, which concentrate higher maintenance costs and lower relative availability.

7.2 Technology Impact on Farm Performance

Model	Turbines	Nominal Capacity	Observed Effect
V100 (2023)	T001–T005	2,000 kW	Low to medium CF, higher corrective rate
V110 (2024)	T006–T007	2,200 kW	Performance improvement without increasing availability
GE 2.5-116 (2025)	T008–T010	2,500 kW	Significant CF increase ($\approx +10$ points) with controlled costs

7.3 Technical and Business Recommendations

7.3.1 Improvement Strategy for V100 Turbines: Integrated Preventive–Predictive Maintenance

The trend observed in V100 and V110 turbines indicates that a large part of corrective events comes from failures in predictable subsystems (pitch, yaw, mechanical vibration, lubrication). Since corrective costs directly impact availability, a sensor-based and early

warning approach would allow anticipating failures and extending the useful life of critical components.

The analysis of Vestas V100 turbine performance (particularly T003 and secondarily T001 and T004) shows an imbalance between energy generated and maintenance costs:

- Lower relative generation compared to turbines in the same group.
- Higher frequency of corrective interventions.
- Higher specific cost per MWh above park average.
- Marked variability in monthly availability.
- Increased mechanical stress from repetitive failures in critical subsystems.

Proposed Actions:

- Increase frequency of scheduled inspections — historical failure data shows recurrence in predictable subsystems (pitch, yaw, gearbox).
- Standardize V100-specific checklists to reduce variability and prevent small anomalies from escalating into complex corrective maintenance.
- Schedule lubrication and adjustments based on equivalent operating hours, not fixed intervals.
- Incorporate predictive monitoring tools: vibration sensors in tower, nacelle, and bearings; thermal monitoring in generator; online oil analysis; SCADA signal validation for micro-shutdown tracking.

Expected Impact:

- 20–30% reduction in corrective maintenance events, shifting unexpected failures toward planned interventions.
- Direct reduction in inactive hours, with proportional annual availability improvement.
- Extended useful life of critical components, reducing emergency intervention costs.
- Stabilization of availability and monthly performance, reducing operational variability of the V100 group.

Conclusion: Integration of preventive and predictive maintenance constitutes a low-CAPEX strategy that allows significant improvement of V100 turbine performance, reducing the gap with technologically newer units without requiring replacements.

7.3.2 Technological Standardization Toward GE 2.5-116 Models

The most recently installed turbines (GE 2.5-116) evidence:

- Greater generation per installed MW
- Lower rate of recurring failures
- Lower corrective costs
- Better energy/operational cost ratio

From a strategic perspective, advancing toward a more homogeneous fleet would enable:

- Simplifying spare parts inventories
- Standardizing maintenance protocols
- Obtaining better agreements with suppliers
- Reducing the levelized cost of energy (LCOE)

Recommendation: Prioritize the GE 2.5-116 model for future expansions and consider its adoption as the park's technological standard. Adopting this model as the base technology increases energy generated per installed MW, reduces failure frequency, and stabilizes maintenance costs.

7.3.3 Optimization of Planned Shutdown Scheduling

Monthly generation data shows wind resource peaks during May–September, while months of lower wind concentrate between November and February. This enables a clear strategy:

- Relocate major maintenance and planned shutdowns to low-wind months
- Minimize unavailability during peak season
- Leverage stable meteorological windows for critical elevated work

Expected result: An additional reduction in losses due to unavailability, increasing captured energy without additional investments.

7.4 Relevant Technical Cases

7.4.1 Technical Case – Turbine T003 (Critical)

Context and Justification

From the wind farm reliability analysis, turbine T003 is consistently identified as the asset with the greatest negative impact in terms of maintenance cost and relative availability. T003 recurrently appears at the unfavorable extreme of the maintenance cost vs. availability correlation, as the unit with highest accumulated cost and lowest relative availability in the park.

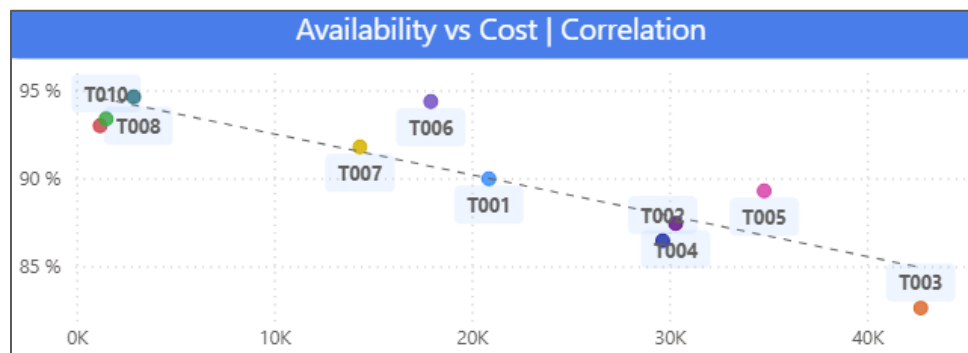


Figure 5. Availability vs cost correlation.

Maintenance and Reliability Behavior

The historical intervention analysis shows T003 concentrates a high number of corrective maintenance events across all three analyzed years (2023–2025), with recurring events year after year. Unlike other park turbines, T003's corrective events do not appear as isolated occurrences but as a repetitive pattern.

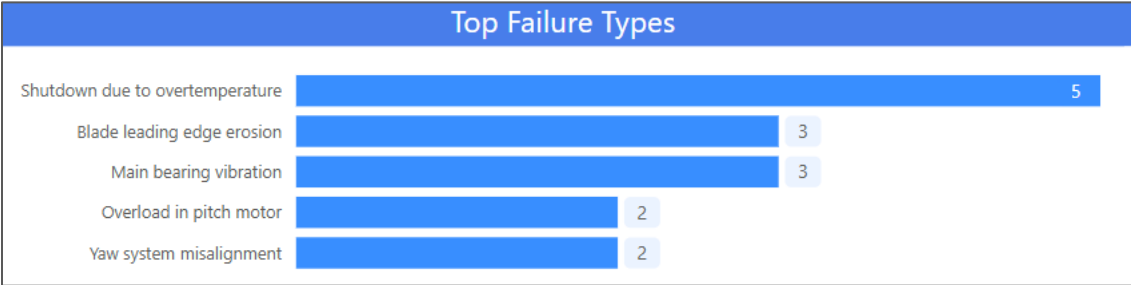


Figure 6. T003 Top Failures

The failure type analysis shows that the largest proportion of corrective events in T003 is associated with overtemperature shutdowns, followed by pitch motor overload, main bearing vibration, yaw system misalignment, and blade leading edge erosion.

Economic and Operational Impact

T003 presents the highest accumulated maintenance cost in the park, with a disproportionate weight relative to its contribution. Despite interventions performed, the turbine maintains below-average availability levels, indicating that maintenance spending has limited operational return in terms of reliability.

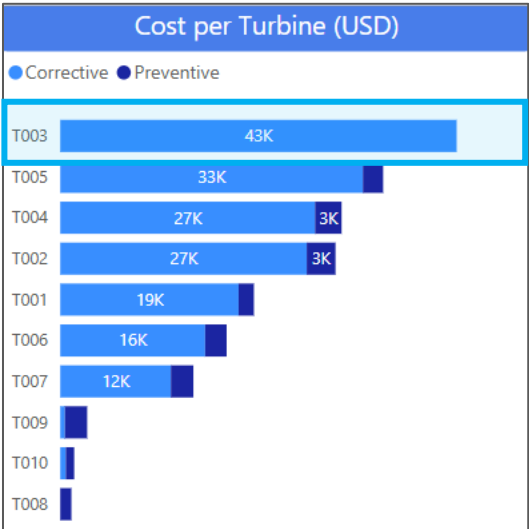


Figure 7. Cost per wind turbine (USD).

Technical Recommendations:

- Prioritize T003 for deep technical analysis focused on thermal management and control subsystems.
- Implement specific predictive maintenance using temperature and vibration monitoring to anticipate failure conditions.
- Evaluate structural interventions (cooling improvements, control adjustments, critical component replacement) oriented toward reducing corrective recurrence.
- Analyze the economic viability of a retrofit or major intervention, considering recurring maintenance cost versus expected benefit in availability.

7.4.2 Technical Case – Turbines T008/T009/T010 (Benchmark)**Context and Justification**

In contrast with critical turbines, T008, T009, and T010 stand out within the wind farm for their stable operational behavior, low maintenance cost, and high relative availability. Their consistent performance positions them as an internal benchmark, useful for identifying replicable good technical and operational practices.

Maintenance and Reliability Behavior

The maintenance history of T008, T009, and T010 shows a clear predominance of preventive interventions, with low corrective maintenance incidence. The failure type Pareto analysis confirms these turbines do not appear relevantly among the main contributors to corrective events in the park.

Economic and Operational Impact

T008, T009, and T010 present some of the lowest maintenance costs in the park, consistently remaining in the favorable zone of the maintenance cost vs. operational availability correlation. This combination reflects high efficiency in maintenance budget utilization, with low relative spending and high operational return.

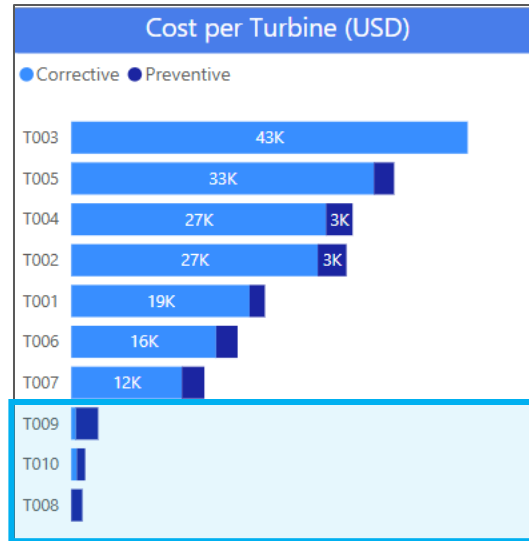


Figure 8. Cost per wind turbine (USD).

Lessons Learned and Replicability:

- Greater emphasis on preventive maintenance reduces the need for costly corrective interventions.
- Operational stability directly translates into lower cost per unit of energy generated.
- The combination of appropriate technology and coherent maintenance strategy generates sustained benefits over time.

The comparison between the critical case (T003) and the benchmark case (T008) demonstrates that improvement in overall farm performance does not depend solely on increasing interventions, but on aligning technology, maintenance strategy, and operating conditions.

7.5 General Conclusion

Based on the results obtained, lines of action emerge aimed at consolidating the improvements achieved and sustaining farm performance over time. The analysis shows the need for differentiated asset management, focusing technical actions on turbines with recurring low performance such as T003, whether through specific adjustments, advanced maintenance strategies, or medium-term replacement evaluations.

In parallel, the positive evolution associated with greater weight of preventive maintenance confirms the advisability of formalizing this approach, using historical information to optimize maintenance planning and budget. The seasonality of wind resource additionally offers an opportunity to reschedule non-critical interventions during lower generation periods, minimizing impact on annual energy produced.

Finally, evidence supports technological standardization toward more efficient models, such as GE 2.5-116 turbines, as a criterion for future expansion or renewal decisions, reinforcing a more efficient, predictable, and economically sustainable operational scheme.

Together, these actions allow moving from a reactive to a proactive, data-driven approach, where technical and economic decisions are supported by objective performance, reliability, and cost indicators. This framework facilitates investment prioritization, continuous operational improvement, and greater predictability of the park's economic result, strengthening its competitiveness and adaptability to future growth scenarios or changes in resource conditions.